Certified High-Precision Equilibria for Poker Subgames via Support-Restricted Proximal Newton Methods

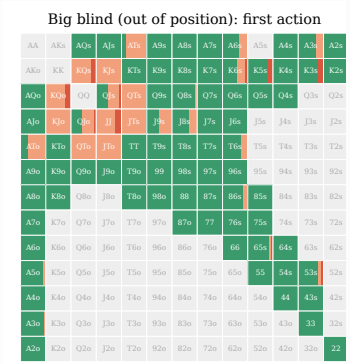
Numerical Analysis Research Club

Background: computational poker, 2015 to 2019

Cepheus (2015) essentially solved heads-up limit hold'em. DeepStack and Libratus (2017) defeated professional players at heads-up no-limit hold'em, and Pluribus (2019) did so with six players.

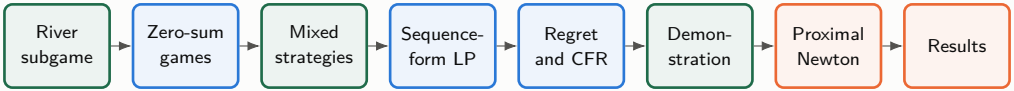
Commercial solvers now compute equilibrium strategies for user-specified subgames.

All of these systems are built on counterfactual regret minimization (CFR). We analyze CFR as a numerical method and propose a second-order refinement.



Equilibrium strategy for the big blind's first action, computed with our solver. Cells are hand classes and colours are actions.

Outline



A poker equilibrium is the solution of a large, structured bilinear saddle-point problem. First-order methods such as CFR are inexpensive per iteration but converge sublinearly.

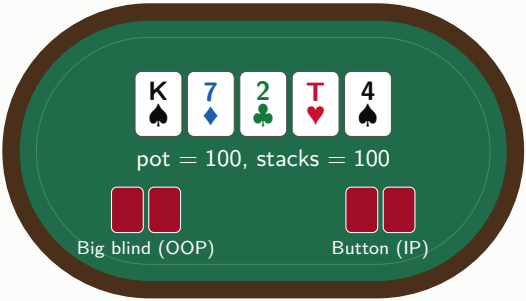
We demonstrate a river subgame solver, then present a proximal Newton refinement that reaches machine precision with a best-response certificate.

Heads-up no-limit hold'em

Each player receives two private cards. Five public cards, the *board*, are dealt in three stages: flop, turn and river.

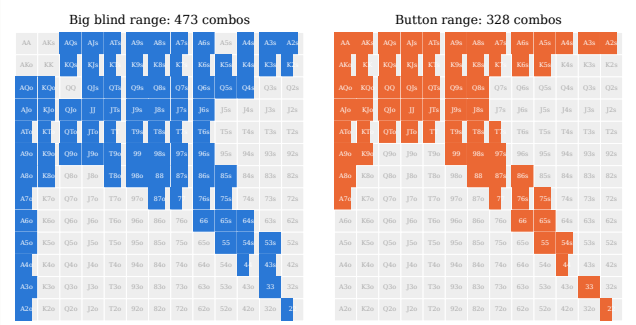
A betting round follows each stage. The actions are check, bet, call, raise and fold, and in no-limit play a bet may be any amount up to the stack.

At showdown the best five-card hand from seven cards wins the pot.



Running example: a river subgame.
The big blind acts first (out of position).

Ranges



A **range** is a probability vector over the 1326 two-card hands, of which 1081 are consistent with the board.

The two ranges differ because the button raised and the big blind called before the flop.

A solver returns an action distribution for every hand at every decision point.

Zero-sum games, Nash equilibrium and exploitability

Strategies lie in compact convex sets, $x \in X$ and $y \in Y$, and the payoff $x^\top A y$ is bilinear.

Minimax theorem (von Neumann 1928)

$$\max_{x \in X} \min_{y \in Y} x^\top A y \;=\; \min_{y \in Y} \max_{x \in X} x^\top A y \;=:\; v^*.$$

A pair (x^*, y^*) attaining both sides is a **Nash equilibrium**. Each equilibrium strategy guarantees v^* against every opponent.

Exploitability

$$\mathrm{expl}(x, y) \;=\; \tfrac{1}{2}\Big(\max_{x'} x'^\top A y \;-\; \min_{y'} x^\top A y' \Big) \;\geq 0,$$

with equality if and only if (x, y) is an equilibrium.

Mixed strategies and the indifference principle

Toy river with pot 1. The bettor holds the nuts or air and bets s. The caller holds a bluff-catcher.

Indifference of the caller fixes the bluff fraction β, and indifference of the bluffing hands fixes the calling frequency c, the *minimum defence frequency* (MDF):

$$\beta = \frac{s}{1+2s}, \qquad c = \frac{1}{1+s}.$$

Every pure strategy is exploitable. Each player mixes so that the opponent is indifferent.

bet s	bluffs β	MDF
1/3	20%	75%
3/4	30%	57%
1	33%	50%
2	40%	33%

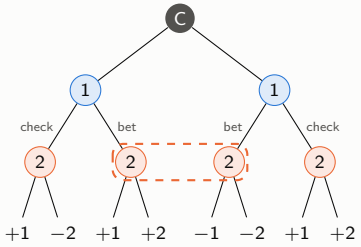
The MDF column is compared with the solver below.

Extensive-form games with imperfect information

Chance deals the cards, the players act in turn, and the leaves carry payoffs.

Nodes that a player cannot distinguish form an **information set**. A behavioural strategy assigns one action distribution to each information set.

game	information sets
Kuhn poker	12
Leduc hold'em	288
Limit hold'em	3.2×10^{14}
No-limit hold'em	about 10^{161}



Player 2 observes the bet but not the deal, so the boxed nodes form one information set.

The sequence-form linear program

A *realization plan* assigns to each sequence σ of player 1's actions the product x_σ of her action probabilities along σ. Realization plans satisfy the flow constraints $Ex = e$, $x \geq 0$, and the expected payoff is bilinear, $u(x, y) = x^\top A y$.

Theorem (Koller, Megiddo and von Stengel 1996)

An equilibrium strategy of player 1 solves

$$\max_{x,\, q} \; f^\top q \quad \text{subject to} \quad A^\top x - F^\top q \geq 0, \quad Ex = e, \quad x \geq 0,$$

a linear program of size linear in the game tree.

On our river A has 2.9 million nonzeros, since each leaf contributes a dense hand-by-hand block. HiGHS did not terminate within 25 minutes. DCFR reaches 1.3×10^{-4} chips in 16 seconds.

Regret matching

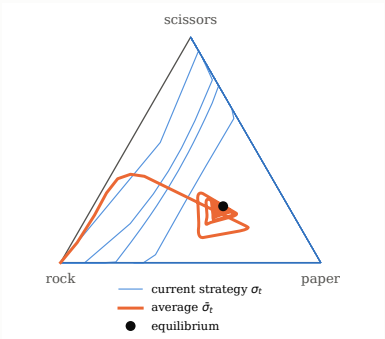
Regret matching plays each action in proportion to its positive cumulative regret.

$$R_T(a) = \sum_{t=1}^{T} \big(u_t(a) - u_t(\sigma_t)\big), \quad \sigma_{T+1}(a) = \frac{[R_T(a)]^+}{\sum_b [R_T(b)]^+}.$$

Theorem (Hart and Mas-Colell 2000)

$\max_a R_T(a) \le \Delta\sqrt{|A|\,T}$, where Δ is the payoff range.

In self-play on a zero-sum game the *average* strategies converge to an equilibrium.



Weighted rock-paper-scissors. The iterate cycles and the average converges.

Counterfactual regret minimization

CFR runs regret matching independently at each information set I. The payoff is the **counterfactual value**, weighted by the probability that chance and the opponent reach I:

$$v_i(\sigma, I, a) = \sum_{h \in I} \pi^{\sigma}_{-i}(h) \sum_{z \sqsupset ha} \pi^{\sigma}(ha \to z)\, u_i(z).$$

Theorem (Zinkevich, Johanson, Bowling and Piccione 2007)

The total regret is bounded by the sum of the local counterfactual regrets. Hence

$$\mathrm{expl}(\bar{x}_T, \bar{y}_T) \;\leq\; \frac{\Delta\, |\mathcal{I}| \sqrt{|A|}}{\sqrt{T}}.$$

Each iteration is one tree traversal, and A is never formed.

Range-vectorized CFR on the public tree

Betting is public and only the cards are private. Each public node stores vectors indexed by hand, such as the reach probabilities π_1 and π_2, and leaf values are matrix-vector products:

$$v_1 = \text{amount} \cdot B\,\pi_2, \qquad B_{ij} = \underbrace{C_{ij}}_{\text{no shared card}} \cdot \begin{cases} 1 & \text{after a fold} \\ \operatorname{sign}(s_i - s_j) & \text{at a showdown} \end{cases}$$

Here s_i is the strength of hand i. After sorting by strength the sign matrix is $+1$ below the diagonal and -1 above it, and C is the all-ones matrix minus a sparse card-removal correction.

Hence $B\pi$ costs $O(n \log n)$ operations instead of $O(n^2)$.

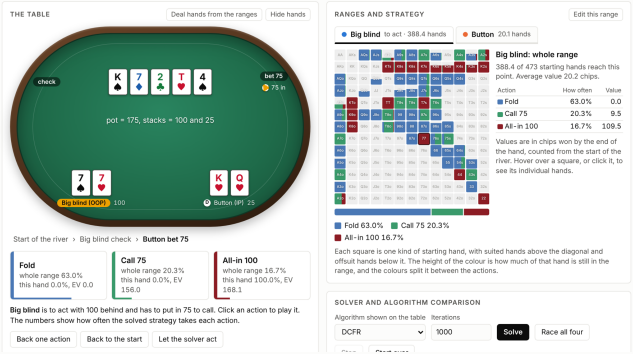
CFR$^+$, DCFR and PCFR$^+$

Modern variants modify the regret update and the averaging weights.

method	modification	interpretation
CFR$^+$	Regrets clipped at zero, linear averaging weights.	Projection onto the nonnegative orthant.
DCFR	Regrets and averages discounted, $(\alpha, \beta, \gamma) = (1.5, 0, 2)$.	Forgetting factors.
PCFR$^+$	Strategy computed from the regrets plus a predicted regret.	Optimism, as in extragradient methods.

All variants update the players alternately (Gauss–Seidel). All share the $O(T^{-1/2})$ bound, and the modern variants are much faster in practice.

Demonstration: a river subgame solver



Big blind facing a 75-chip bet. Blue: fold. Green: call. Red: all-in.

`app/poker-solver.html` solves the river subgame in the browser. We examine:

1. the convergence of CFR, CFR$^+$, DCFR and PCFR$^+$,
2. the big blind's strategy over its 473 hands,
3. the ranges along the game tree,
4. action values of mixed hands,
5. agreement with the minimum defence frequency.

The board, ranges and bet sizes are adjustable.

Equilibrium structure and minimum defence frequency

The equilibrium is polarized. Strong hands and some weak hands bet, and medium hands check.

Mixed hands have equal action values, as complementary slackness requires.

Against the big blind's bets, defence is within 1.5 points of the MDF. Against the button's bets and against all-in raises it is far lower, because there the bluffing hand's alternative action has positive value.

defender	line	s	defends	MDF
Button	BB checks, BTN bets 75, BB jams	0.10	64.7%	90.9%
Button	BB bets 33	0.33	74.3%	75.2%
Button	BB bets 75	0.75	55.8%	57.1%
Button	BB bets 100	1.00	49.1%	50.0%
Big blind	BB bets 75, BTN jams	0.10	63.4%	90.9%
Big blind	BB checks, BTN bets 33	0.33	52.2%	75.2%
Big blind	BB bets 33, BTN jams	0.40	56.3%	71.2%
Big blind	BB checks, BTN bets 75	0.75	37.5%	57.1%
Big blind	BB checks, BTN bets 100	1.00	32.6%	50.0%

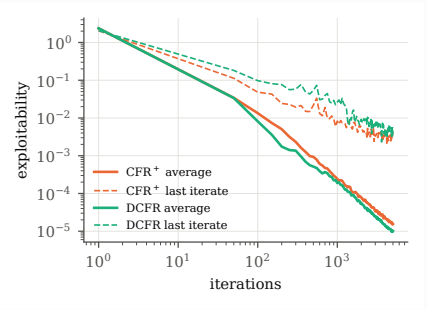
Exact equilibrium. s is the bet as a fraction of the pot, $\mathrm{MDF} = 1/(1+s)$, and a jam is an all-in bet.

Motivation for high-precision solutions

Practical solvers stop near 0.1% of the pot.

Higher precision is needed for repeated real-time re-solving, for identifying the exact support, and for certified reference solutions.

CFR is a first-order method on a nonsmooth, degenerate problem, and its tail is sublinear. The standard remedy is a locally superlinear method started from a good initial point.



Leduc. Only the average converges, and each additional digit costs about ten times more iterations.

The equilibrium as a skew-symmetric variational inequality

Let $z = (x, y, p, q)$ collect both realization plans and both vectors of information-set values. Equilibria are exactly the solutions of the optimality conditions of both linear programs:

$$0 \le x \;\perp\; E^\top p - Ay \;\ge 0, \qquad Ex = e,$$
$$0 \le y \;\perp\; A^\top x - F^\top q \;\ge 0, \qquad Fy = f.$$

These conditions have the form $w = Mz + b$ with a **skew-symmetric** matrix:

$$M = \begin{pmatrix} 0 & -A & E^\top & 0 \\ A^\top & 0 & 0 & -F^\top \\ -E & 0 & 0 & 0 \\ 0 & F & 0 & 0 \end{pmatrix}, \qquad M + M^\top = 0.$$

The problem is therefore monotone, but its solution set is a polyhedron rather than a point, so Newton's method does not apply directly. CFR provides an initial point for all of z.

Proximal point regularization

Given z^k, the iterate z^{k+1} solves the same conditions with

$$w_\rho(z) = (M + \rho I)\, z + b - \rho\, z^k.$$

Since $d^\top (M + \rho I) d = \rho \|d\|^2$, each subproblem is strongly monotone and has a unique solution.

Theorem (Rockafellar 1976, Luque 1984). The iterates converge for every $\rho > 0$. For polyhedral problems the convergence is linear, with a rate that tends to zero as $\rho \to 0$.

Small ρ gives fast outer convergence and ill-conditioned subproblems.

Leduc, $\rho = 10^{-5}$

Residual after each outer step:

$3.4 \times 10^{-2} \to 4.8 \times 10^{-6} \to 3.3 \times 10^{-9}$
$\to 6.9 \times 10^{-12} \to 2.0 \times 10^{-14}$

About three digits per step.

Semismooth Newton and the primal-dual active-set method

Complementarity is equivalent to the piecewise-linear equation $\min\big(x_i,\ w_\rho(z)_i\big) = 0$.

A semismooth Newton step fixes a guessed active set (the dropped actions), imposes indifference on the remaining actions and solves one linear system with matrix $V = D_a + D_b(M + \rho I)$, where $D_a, D_b \geq 0$ are diagonal.

Lemma. Every such V with $D_a + D_b$ positive definite is nonsingular.

Suppose $Vd = 0$ and put $u = (M + \rho I)d$. Row i reads $(D_a)_{ii} d_i + (D_b)_{ii} u_i = 0$ with nonnegative coefficients that are not both zero, so $d_i u_i \leq 0$. Hence $\rho\|d\|^2 = d^\top u = \sum_i d_i u_i \leq 0$, and $d = 0$. □

A correct active set gives the *exact* solution in one step. Convergence is only local, so a good initial guess is required.

Interior-point crossover and quasi-definite factorization

Initialization. The first subproblem is solved by a Mehrotra interior-point method from the CFR point. Its solution predicts the active set, and active-set steps then terminate exactly (crossover). Later subproblems need one to three linear solves.

Linear algebra. After sign changes in two row blocks every system is symmetric *quasi-definite*. By Vanderbei's theorem an $LDL^\top$ factorization exists for every symmetric ordering, so a fill-reducing ordering can be fixed in advance.

One factorization, in seconds

	small river	our river
general sparse LU	0.11	25
quasi-definite $LDL^\top$	0.04	7.6

A residual of 10^{-13} from the CFR point requires 19 linear solves on Leduc and 17 on the small river.

Support restriction and best-response verification

On larger instances the solution set is *flat* along unplayed actions and unreached information sets. A small residual then does not imply a small error, and the predicted active set is wrong.

We restrict the problem to the actions that DCFR plays with probability at least 10^{-3} and solve the restricted problem exactly.

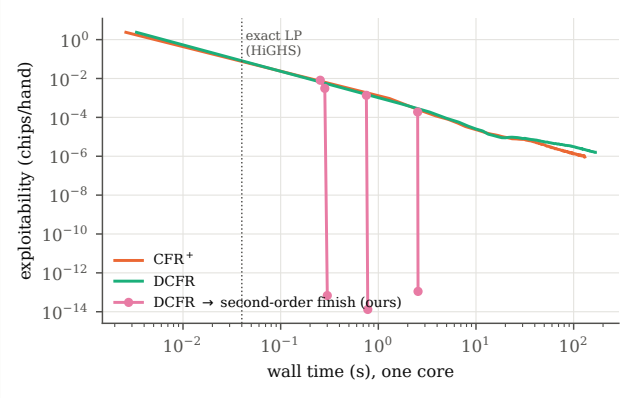
Pruned actions with positive counterfactual regret in the full game are restored, and the process repeats.

Algorithm

1. Run DCFR for a few thousand iterations.
2. Restrict to its support.
3. Solve the restricted problem to machine precision.
4. Compute exact best responses in the full game.
5. Restore profitable actions and return to 3.

Step 4 certifies the final exploitability in the full game.

Results: Leduc hold'em

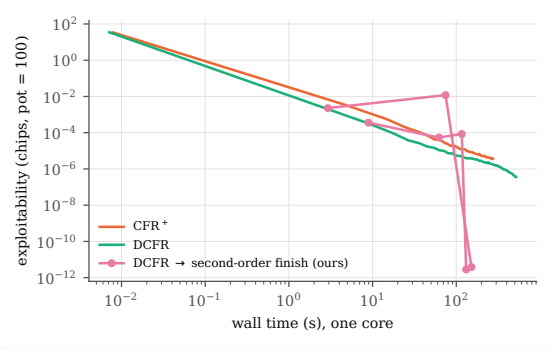


Each pink curve starts from a DCFR warm start.

From 100 DCFR iterations the method reaches 6.8×10^{-14} in 0.30 seconds. CFR^{+} reaches 9.2×10^{-7} after 130 seconds.

HiGHS solves Leduc in 0.04 seconds, so the relevant comparison is the convergence profile.

Results: river subgame with 473 and 328 hands



17,626 unknowns. Points mark verification rounds.

After 3,000 DCFR iterations, 3 rounds on 34% of the unknowns reach 2.8×10^{-12} chips in 131 seconds.

DCFR alone reaches 3.9×10^{-6} in the same time, and 3.6×10^{-7} after 190,000 iterations (520 seconds).

Without support restriction the Newton refinement is slower than DCFR. HiGHS did not terminate within 25 minutes.

Limitations and open problems

Full ranges: 1,081 hands per player, 47,568 unknowns

The restricted problem retains 43% of the unknowns, and one factorization takes about 10 seconds. After 80 minutes the method reaches 4.1×10^{-6} chips, and DCFR reaches 9.5×10^{-6} in 10 minutes. **CFR is faster.**

The bottleneck is factorization. Each payoff block is a sorted sign matrix with a sparse correction and admits $O(n \log n)$ products. A Newton–Krylov method therefore needs a preconditioner for shifted skew systems $(D + M + \rho I)\, d = r$.

Further directions are support prediction from CFR regrets and exact re-solving in real-time agents.

Contributions and related work

The components are classical: the sequence form, the proximal point method, semismooth Newton, interior point with crossover, quasi-definite factorization and CFR.

The closest work is Yang, Gao, Lin and Kroer (2025), a semismooth Newton method for *matrix* games with a first-order warm start.

Our contributions for extensive-form games:

- a skew variational inequality on the sequence form, with proximal regularization and a nonsingularity lemma,
- a primal and dual warm start from CFR,
- restriction to the support predicted by CFR,
- certification by best responses in the full game.

Conclusions

The sequence form reduces equilibrium computation to a linear program of size linear in the game tree.

CFR is regret matching at every information set. It is matrix-free, converges at rate $O(T^{-1/2})$ in theory, and is much faster in practice.

The computed strategies exhibit polarization, slow play and defence consistent with the MDF.

A support-restricted proximal Newton method reaches an exploitability of 10^{-12} in about two minutes on a river with 17,626 unknowns, where DCFR reaches 10^{-6}. On full ranges the factorization cost favours CFR.

Code, benchmarks and the interactive solver: `github.com/dwangnumbers/Poker-Solver`.

References

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The complete list of references is in `talk-detailed.pdf`.

Appendix: non-uniqueness in Kuhn poker

Kuhn poker uses three cards, 

, an ante of one chip and a single bet of one chip.

Its equilibria form a one-parameter family. Player 1 bluffs the jack with probability $\alpha \in [0, \frac{1}{3}]$ and bets the king with probability 3α. The value is $-\frac{1}{18}$.

The solution set is a segment, not a point.

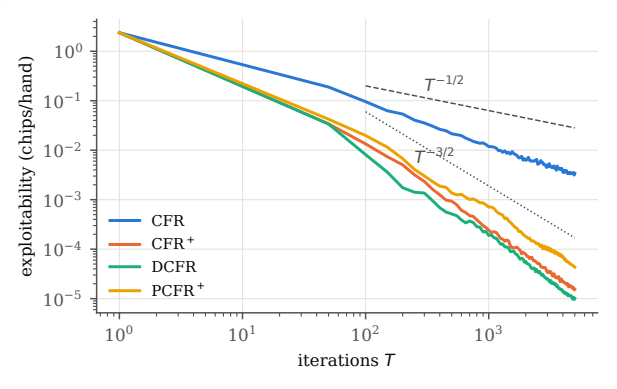
Computed

$\alpha = 0.198$, and the king is bet with probability $0.593 = 3\alpha$.

Value $-0.0555\ldots = -1/18$.

Exploitability 1.1×10^{-16}.

Appendix: first-order convergence on Leduc hold'em

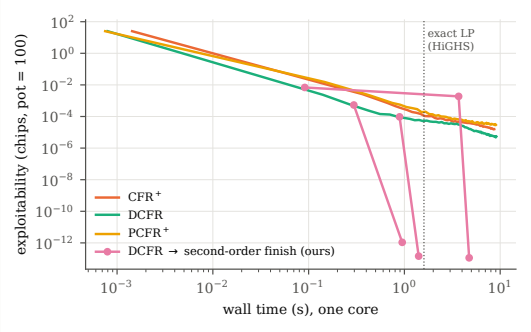


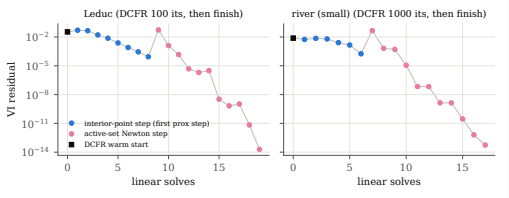
Exploitability of the average strategy.

DCFR reaches 1.0×10^{-5} after 5,000 iterations (14 seconds). After 50,000 iterations (130 seconds) CFR$^+$ reaches 9.2×10^{-7} and DCFR 1.6×10^{-6}.

Each additional digit costs about ten times more iterations.

Appendix: small river and inner iteration history





Left: river with 147 and 139 hands. The method reaches 1.1×10^{-12} chips in 0.95 seconds, DCFR reaches 5.4×10^{-6} after 9 seconds, and HiGHS needs 1.6 seconds. Right: residual after each linear solve. Blue points are interior-point iterations and pink points are Newton steps.

Appendix: proximal point rate and flat solution sets

The proximal step is the resolvent of a maximal monotone operator T, and the iterates are Fejér monotone with respect to the solution set Z.

If T^{-1} is upper Lipschitz at 0 with modulus a, which holds for polyhedral problems, then

$$\operatorname{dist}(z^{k+1}, Z) \;\leq\; \frac{a\rho}{\sqrt{1 + a^2\rho^2}}\; \operatorname{dist}(z^k, Z).$$

The modulus a bounds the distance to Z in terms of the residual. It is large when the solution set is flat.

Fast outer convergence requires $\rho \ll 1/a$, but $\|(M + \rho I)^{-1}\| \approx 1/\rho$ makes the active set of each subproblem sensitive. On Leduc the observed factor is about 10^{-3} per step at $\rho = 10^{-5}$. On the large river no single ρ achieved both, and support restriction removes the flat directions instead.